

B.M.S College of Engineering Bengaluru
Autonomous Institute, Affiliated to VTU



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Report on

PHASE LEAD CIRCUIT AND BODE PLOT

Submitted in partial fulfilment of requirements
of IV semester

ANALOG INTEGRATED CIRCUIT - 23EC4PCAIC

of

BACHELOR OF ENGINEERING in
Electronics and Communication Engineering

Submitted by

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Simulation of Phase Lead Circuit and Bode Plot

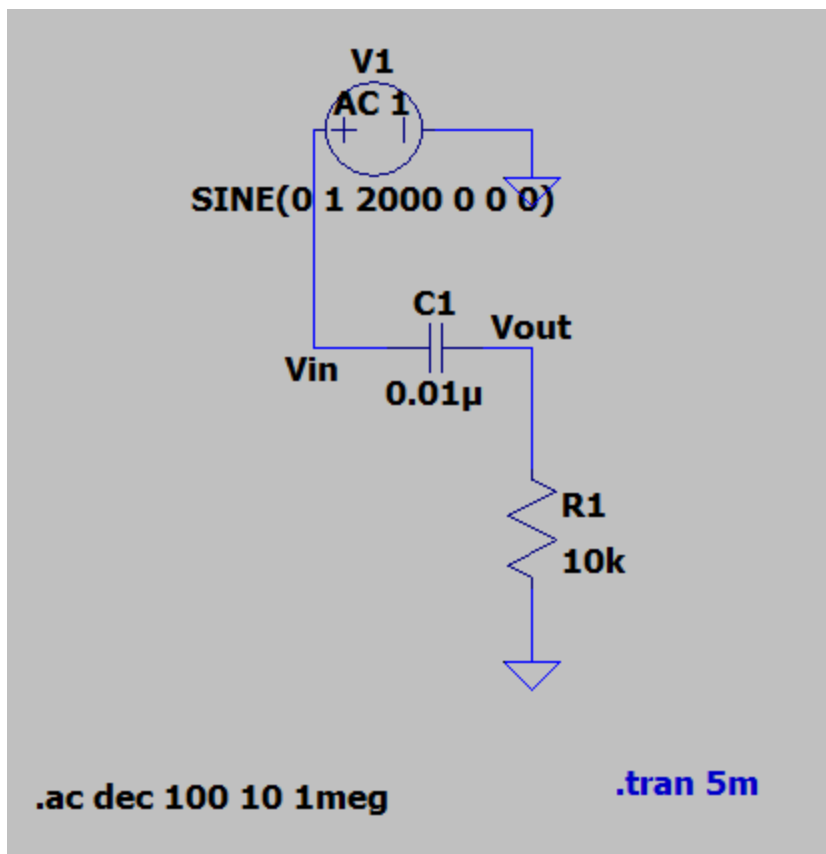
1. Aim

To design and simulate a Phase Lead Circuit and analyze its frequency response using waveform analysis and Bode plots in LTSpice.

2. Software Used

LTspice

3. Circuit Diagram



4. Components Used

Components	Value
Resistor	10kΩ
Capacitor	0.01uF
AC source	1V

5. Input Signal

Sine Wave Input

Parameter	Value
DC Offset	0V
Amplitude	1V
Frequency	2kHz

5. Theoretical Background

A Phase Lead Circuit is a first-order RC network that provides positive phase shift between input and output.

Transfer Function:

$$H(s) = \frac{sRC}{1 + sRC}$$

Cutoff Frequency:

$$f_c = \frac{1}{2\pi RC}$$

Calculation :

$$f_c = \frac{1}{2\pi RC}$$

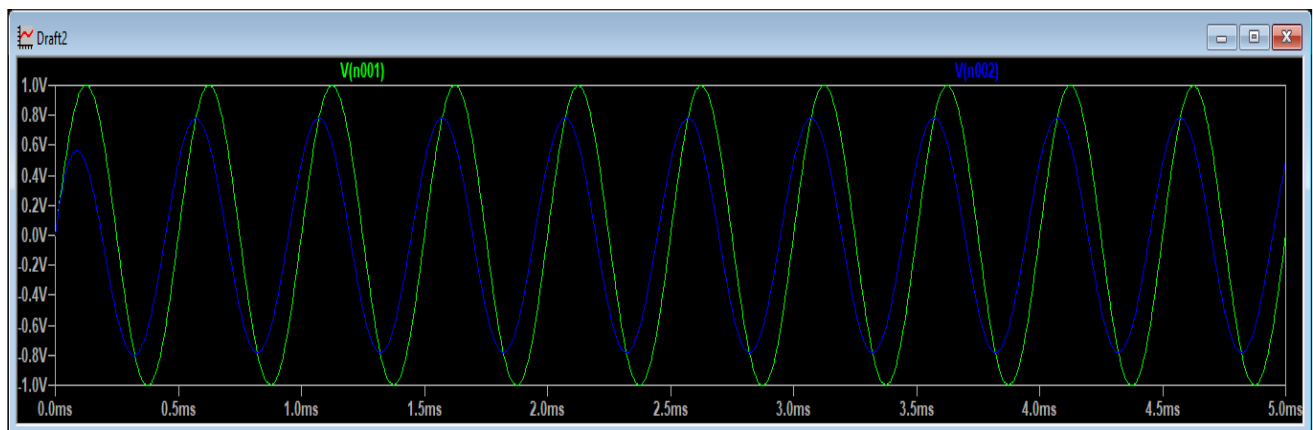
$$R = 10k\Omega$$

$$C = 0.01\mu F$$

$$f_c = \frac{1}{2\pi(10k)(0.01\mu)}$$

$$f_c = 1591Hz$$

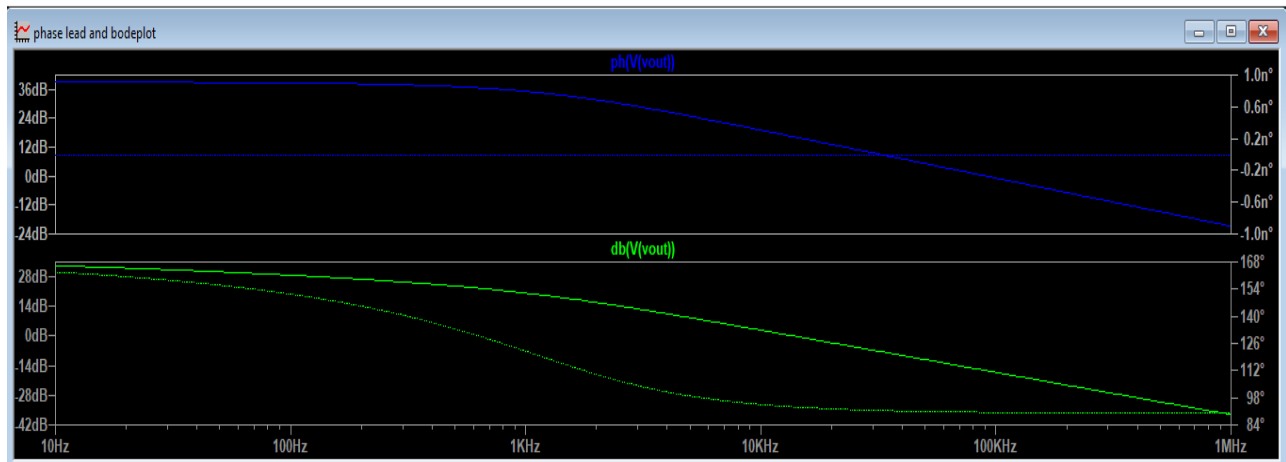
6. Waveform Analysis



Observation:

- Input and output signals are sinusoidal.
- Output waveform leads the input waveform in phase.
- The circuit behaves as a phase lead network

7. Bode Plot



Observation:

Magnitude Response

- Gain varies with frequency.
- The circuit attenuates low-frequency signals more strongly.
- High-frequency signals experience lower attenuation.

Phase Response

- The output leads the input in phase.
- Phase shift changes with frequency.
- The phase lead is maximum near the cutoff region.

8. Self-Learning Behaviour

The phase lead circuit demonstrates adaptive behaviour through its frequency-selective response.

- Low-frequency signals are attenuated.
- High-frequency signals are passed effectively.
- The RC network adjusts its response depending on input frequency.
- Changing resistor and capacitor values modifies circuit behaviour, similar to weight adjustment in self-learning systems.

9. Result

The Phase Lead Circuit was successfully simulated in LTspice. Waveform analysis and Bode plots verified the expected phase lead characteristics and frequency response.